



THERMAL HYDRAULIC INVESTIGATIONS ON LEAD-BISMUTH COOLED FUEL ASSEMBLIES WITH DUCTS

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ABSTRACT

Compared to sodium cooled fast reactors, heavy liquid metal (lead or lead-bismuth) cooled fast reactors use relatively open fuel lattices without wire spacers. In the present work, a subchannel analysis was performed for lead-bismuth cooled fuel assemblies with ducts and thermal hydraulic characteristics were investigated with emphasis on turbulent mixing between subchannels and interassembly heat transfer. The existing SLTHEN code, which had been originally developed for sodium cooled fast reactors, was modified and applied to typical lead-bismuth cooled fuel assemblies with ducts. The analysis started from a single fuel assembly to assess the effect of turbulent mixing and heat transfer between subchannels and was extended to 7 fuel assemblies to investigate interassembly heat transfer. The results of the modified SLTHEN show that the maximum cladding temperature is not largely affected by turbulent mixing and interassembly heat transfer and the simplified analysis such as single assembly consideration is very useful for thermal hydraulic analysis and design of lead-bismuth cooled fuel assemblies under conceptual design stages.

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1. INTRODUCTION

The use of liquid lead or lead-bismuth alloys as a coolant in nuclear reactors has been given a lot of attention by the scientific community in recent years (Spencer, 2000). The attention was boosted after the disclosure of successful Russian experiences related to lead-bismuth cooled nuclear submarines. Compared to pure lead, lead-bismuth alloys are preferred due to their low melting point. The research related with lead or lead-bismuth cooled nuclear reactors has been focused mainly on

two types of reactor concepts, (1) Accelerator-Driven System (ADS) for transmutation of long-lived nuclear wastes and (2) fast reactors for energy productions. In ADS, there is an additional merit that lead-bismuth coolant can also be used as a spallation target. Lots of design concepts for these types of reactors have been explored recently. However, these studies are under initial design stages and most of them were focused on neutronic core design without detailed thermal hydraulic calculations in the fuel rod bundles. Naturally, progression of core design of nuclear reactors into the final design stage requires accurate calculation of thermal hydraulic behavior of the assemblies. Although thermal hydraulic behavior of sodium cooled fuel rod bundles have been intensively studied by many authors, thermal hydraulic studies related with lead-bismuth cooled fuel rod bundles are very rare. In the present work, a subchannel analysis was performed for lead-bismuth cooled fuel assemblies with ducts and thermal hydraulic characteristics were investigated with emphasis on turbulent mixing between subchannels and interassembly heat transfer.

2. DESIGN FEATURES OF LEAD BISMUTH COOLED FUEL ASSEMBLIES

HYPER (HYbrid Power Extraction Reactor) is the ADS which is being developed by Korea Atomic Energy Research Institute (KAERI) (Park *et al.*, 2002). In the present work, the design parameters of the HYPER core shown in Table 1 are used to investigate thermal hydraulic characteristics of lead-bismuth cooled fuel assemblies with ducts.

Table 1. Design parameters of the HYPER core considered for the present work

Parameter	Values
Core :	
Core thermal power [MWth]	1000
Coolant	Pb-Bi eutectic
System operating temperature [$^{\circ}\text{C}$]	340 - 510 $^{\circ}\text{C}$
Cooling type	forced convection
Fuel assembly :	
Number of assembly	237
Assembly pitch [cm]	16.13
Interassembly gap thickness [cm]	0.3
Duct inside flat to flat distance [cm]	15.01
Duct wall thickness (cm)	0.26
Rods per assembly	217
Nominal linear power generation [W/m]	12152.6
Nominal assembly mass flow rate [kg/s]	173.6 kg/s
Spacer type	bare rod
Fuel rod :	
Fuel rod arrangement	triangular
Active height (cm)	160
Outer diameter (cm)	0.67
Pitch/diameter	1.48

The design characteristics of HYPER are similar to those of existing sodium cooled fast reactors in many ways. HYPER uses triangular fuel rod lattices and hexagonal ducts. However the usage of an open lattice ($P/D = 1.48$) requires grid spacers rather than wire spacers which are widely adopted in existing sodium cooled fast reactors. In the present work, bare rod conditions are assumed for the

analysis to simplify the works. The effects of grid spacers on turbulent mixing and heat transfer between subchannels should be considered in future works. They could be small because the property change of liquid lead-bismuth with pressure is very small.

Due to the nuclear properties of lead-bismuth, rectangular fuel lattice can also be considered for critical fast reactors but triangular lattice may be better for ADSs because of the central target zones. One of the other big design alternatives of HYPER is fuel assemblies without ducts. Ductless fuel assemblies have some merits as well as problems of less experience in liquid metal cooled fast reactors. In the present work, only fuel assemblies with ducts are focused.

3. SUBCHANNEL ANALYSIS AND RESULTS

A subchannel analysis code named SLTHEN (Yang, 1997) was used in the present work after slight modifications. The SLTHEN code is based on the ENERGY model (Khan *et al.*, 1975) specifically developed for sodium cooled wire-wrapped fuel rod assemblies with a simplified energy equation mixing model. The previous version of SLTHEN is SE2-ANL (Yang and Yacout, 1995). SE2-ANL is a multi-assembly, steady-state subchannel analysis code for forced convections and it was successfully applied to thermal hydraulic core design and performance analyses of sodium cooled reactors.

In order to use the SLTHEN code for lead-bismuth cooled fuel assemblies without wire spacers, the properties of Pb-Bi were added and pressure drop correlations, flow spilt and turbulent mixing models were modified with bare rod conditions.

3.1 Case 1: Single Assembly

At first, single fuel assembly was considered to examine the applicability of the modified SLTHEN code to lead-bismuth cooled fuel assemblies without wire spacers and to assess the coolant mixing and heat transfer between subchannels in open fuel lattices with ducts. Axial power profile was assumed as chopped cosine shape having a peaking factor of 1.2. Uniform radial power profile was assumed across the assembly. Single fuel assembly of HYPER was modeled as 438 subchannels including interior, edge, corner channels for SLTHEN calculations as shown in Fig. 1. The SLTHEN calculations were performed and the results were obtained in a short computing time.

The summarized results of the SLTHEN calculations are shown in the second column of Table 2. For comparison, subchannel analysis was performed using the conventional subchannel analysis code MATRA (Yoo *et al.*, 1999) and the results are provided in the third column of Table 2.

Table 2. The summary of the subchannel analysis results of the HYPER fuel assembly

	SLTHEN	MATRA
Velocity at channel inlet [m/s]		
Average	1.433	1.421
Interior	1.449	1.432
Edge	1.382	1.389
Corner	1.097	1.176
Pressure drop [kPa]	32.6	34.0
Average exit coolant temp. [°C]	510.0	512.0
Peak coolant temperature [°C]	524.2	524.1
Peak clad temperature [°C]	536.7	536.2

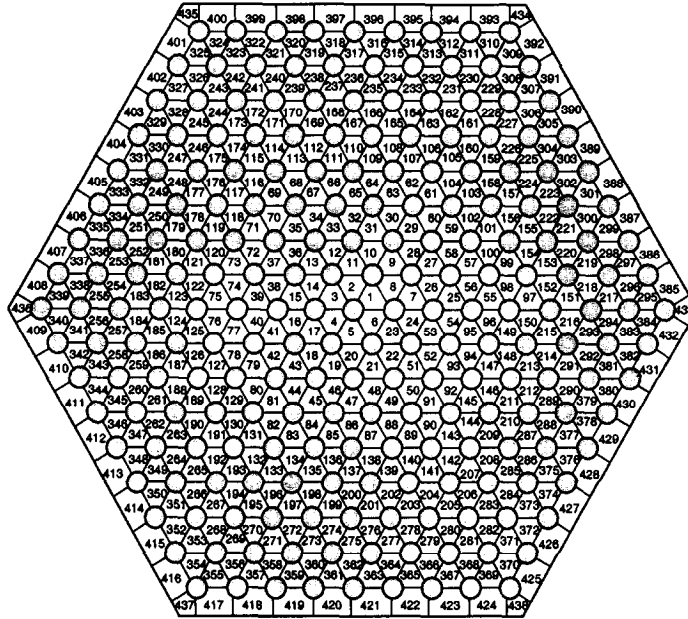


Fig. 1 Subchannels of single fuel assembly of HYPER.

Although the SLTHEN code uses a simpler model, there are good agreements between major results. Comparable accuracy is obtained in a very short computer time with the SLTHEN code. The coolant temperature distributions along the axial height are provided in Fig. 2. The bundle averaged coolant temperature can be analytically calculated since the axial power profile is given as a chopped cosine shape. Figure 2 also shows the adequacy of prediction by the modified SLTHEN code.

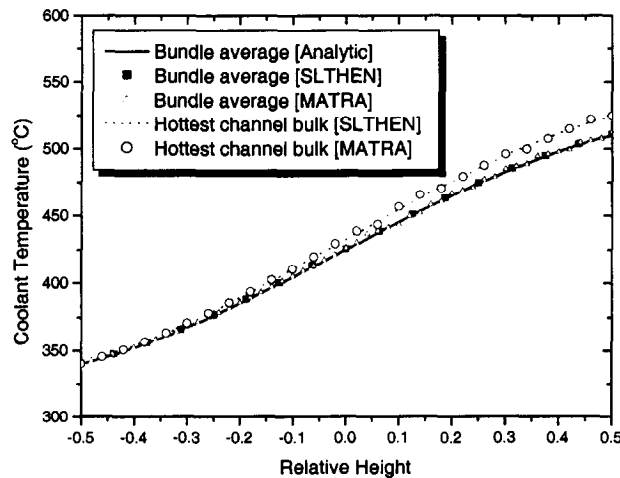


Fig. 2 Coolant temperature distributions along the height.

In Fig.2, the difference between the bundle averaged and the hottest subchannel bulk values of the coolant temperature is less than 15 °C. Certainly, this value can be affected by the turbulent mixings between subchannels. The turbulent mixing between subchannels is considered as the most uncertain parameter affecting the coolant mixing and heat transfer between subchannels. In the SLTHEN code, the turbulent eddy diffusivity (ϵ_H) is used to model a phenomenon of turbulent mixing. Various prediction models for the turbulent eddy diffusivity in bare rod bundles have been proposed (Nikurades, 1932, Dwyer, 1963, Rogers and Rosehart, 1972, Rogers and Tahir, 1975). The turbulent eddy diffusivities were calculated by these models for HYPER design conditions and are shown in Table 3.

Table 3. Turbulent eddy diffusivity calculated by various models for HYPER conditions

Correlations	Nikuradse	Dwyer	Rogers and Rosehart	Rogers and Tahir
ϵ_H (m ² /s)	6.769×10^{-5}	8.121×10^{-6}	4.0×10^{-5}	1.431×10^{-5}

The SLTHEN calculations were made with a wide range of the turbulent eddy diffusivities to investigate its effect. The results are provided in Fig. 3.

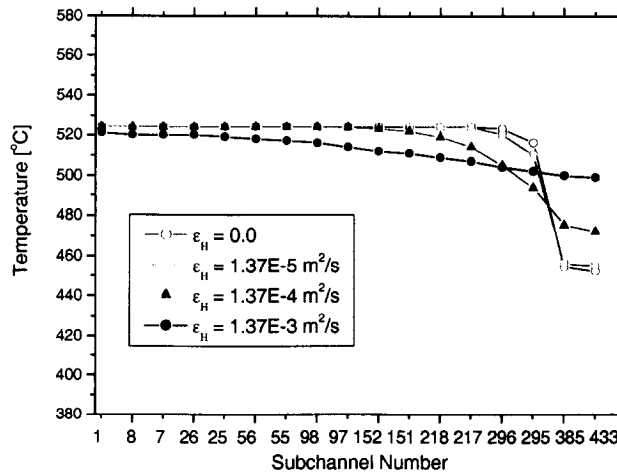


Fig. 3 Radial coolant outlet temperature distribution in single fuel assembly of HYPER.

It can be seen that there is a large temperature difference (~60 °C) within an assembly when there is no turbulent mixing. The temperature gradient between the center and edge channels is reduced with the increase of turbulent mixing. This means that the modified SLTHEN code can reasonably simulate heat transfer between subchannels. As shown in Table 3, the typical value of the turbulent eddy diffusivity for HYPER conditions is expected as $\sim 10^{-5}$ m²/s and only 3 °C of the reduction of the maximum cladding temperature is attained by one hundred times increase of the turbulent eddy diffusivity. Therefore, it can be judged that the maximum cladding temperature within an assembly is not largely affected by turbulent mixing between subchannels so long as HYPER does not adopt mixing promoters such as a mixing vane.

3.2 Case 2: Seven Assemblies

When the fuel assemblies have the ducts, the results of single assembly may be unrealistic since it neglects interassembly heat transfer. It is desirable to understand quantitatively how much the interassembly heat transfer affects the coolant temperature distribution. In order to investigate the effect of heat transfer between assemblies, seven fuel assemblies having three different assembly powers as shown in Fig. 4 were considered.

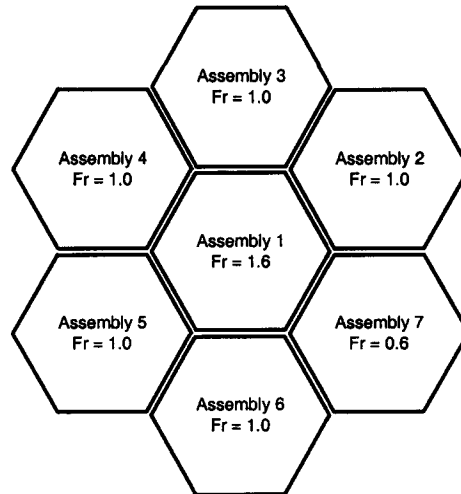


Fig. 4 Seven assemblies considered in the present work.
(Fr is the assembly power peaking factor.)

For axial power profile, chopped cosine shape having a peaking factor of 1.2 is assumed for all assemblies. Before assessing the effect of interassembly heat transfer, the SLTHEN calculation was performed for the seven assemblies shown in Fig. 4 with the assumption of no heat transfer between the assemblies. The results are provided in Table 4(a). It is confirmed that the SLTHEN results without heat transfer between assemblies are the same with those of the single assembly.

Table 4(b) is the result of the calculation considering the heat transfer between assemblies. There are small amounts of changes of the average exit coolant temperatures but there are no changes in the maximum coolant and cladding temperatures.

Table 4. Summary results of SLTHEN calculations for considered seven assemblies

(a) Without heat transfer between assemblies							
Assembly	1	2	3	4	5	6	7
Ave. exit coolant temp.(°C)	612.0	510.0	510.0	510.0	510.0	510.0	442.0
Peak coolant temp. (°C)	634.8	524.2	524.2	524.2	524.2	524.2	450.5
Maximum cladding temp. (°C)	654.7	536.7	536.7	536.7	536.7	536.7	458.0
(b) With heat transfer between assemblies							
Assembly	1	2	3	4	5	6	7
Ave. exit coolant temp.(°C)	609.0	510.1	510.4	510.4	510.4	510.1	443.4
Peak coolant temp. (°C)	634.8	524.2	524.2	524.2	524.2	524.2	450.5

Figure 5 shows the radial coolant outlet temperature distributions of three of the assemblies in Fig. 4 calculated with a wide range of the turbulent eddy diffusivities.

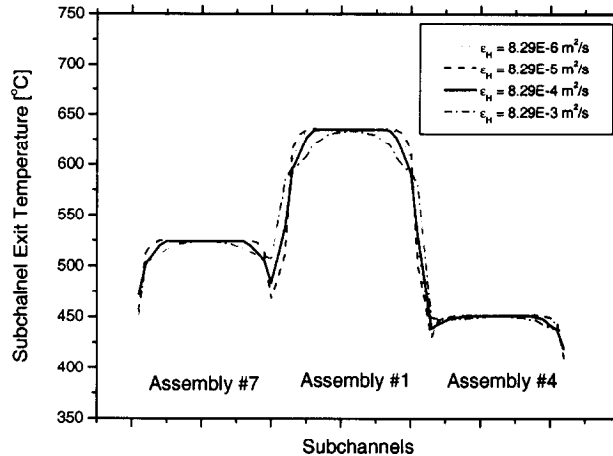


Fig. 5 Radial coolant outlet temperature distributions of three assemblies considered.

Thanks to large temperature gradients at the interfaces of the assemblies, the effect of the turbulent mixing is boosted compare to the single assembly case. It can be seen, however, active heat transfer between subchannels near the interfaces is not propagated enough to decrease the maximum coolant temperatures of the assemblies.

4. CONCLUSIONS

In the present work, the typical fuel assemblies with ducts designed for lead-bismuth cooled ADS were selected and thermal hydraulic characteristics were investigated using the modified SLTHEN code. Two cases, i.e., single and seven assemblies were analyzed to examine the effects of turbulent mixing between subchannels and interassembly heat transfer on the temperature distribution within assemblies with open fuel lattices without wire spacers. The results of SLTHEN show that the maximum cladding temperature, which is an important parameter in thermal hydraulic core design of liquid metal cooled nuclear reactors, is not largely affected by turbulent mixing and interassembly heat transfer. Therefore, the simplified analysis such as single assembly consideration is very useful for thermal hydraulic analysis and design of lead-bismuth cooled fuel assemblies under conceptual design stages.

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